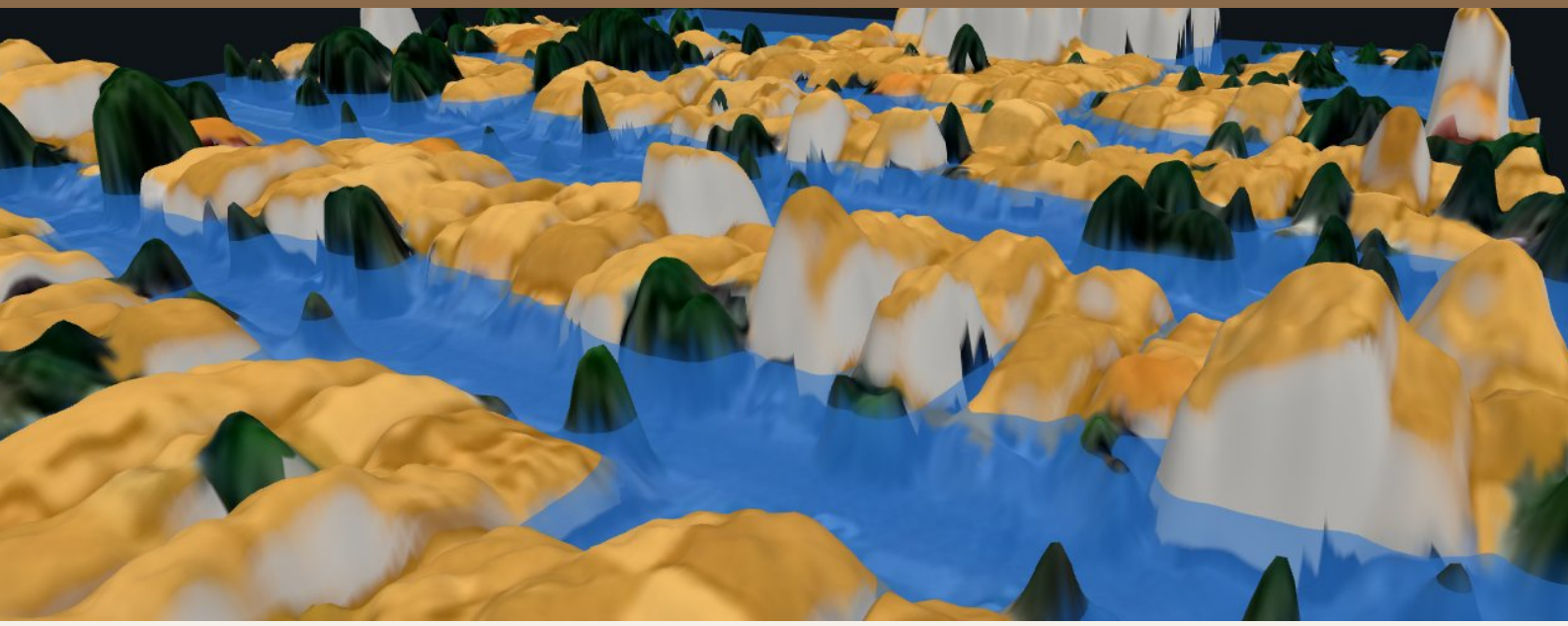
HS202 PROJECT REPORT · IIT ROPAR

# DEPTHWIZARD

One satellite image to heights you can fly through and check



Cover: fly-through of Canoas (Brazil) at the May 2024 flood level, built from one pre-flood image. Orange = roofs above water. Imagery: Maxar Open Data (CC BY-NC 4.0).

| | |
|---|---|
| PROBLEM STATEMENT | SIH26175, “Single-View Height Estimation and 3D Flythrough” Smart India Hackathon 2026 · ISRO · Disaster Management theme |
| COURSE | HS202 · Indian Institute of Technology Ropar |
| TEAM | TerraVista_iitrpr |
| STUDENTS | Himanshu · Shlok |
| DATE | 27 October 2026 |

Every number in this report comes from a row marked “measured” in the project’s claims register (docs/CLAIMS.md), with its 95% confidence interval and a baseline.

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# 2 ABSTRACT

> DepthWizard turns one optical satellite image into a height map of every building, tree and slope, and lets disaster managers fly through it in 3D and check it against reference data.

It is team TerraVista's (IIT Ropar) answer to Smart India Hackathon 2026 problem SIH26175, "Single-View Height Estimation and 3D Flythrough", set by ISRO under the Disaster Management theme. The height model is RS3DAda (Song et al., NeurIPS 2024), used with its released weights and no fine-tuning by us.

For GeoTIFF inputs, DepthWizard turns those heights into an absolute surface model in metres above sea level. It uses the Copernicus GLO-30 terrain model with roofs and trees removed, NASA ICESat-2 laser ground points and optional ground control points. A local, offline app adds a three.js flythrough, in-app validation (MAE, RMSE, correlation) and four disaster tools: Prepare, Respond, Relief and Assess.

Tested zero-shot, the model's height error is 3.90 m [3.24, 4.66] on 150 aerial tiles against airborne LiDAR, where a flat-ground guess scores 5.65 m, and 4.85 m on eight Indian scenes against ICESat-2. It reproduces the published satellite results for Atlanta and Buenos Aires. Its main weakness is that objects taller than 20 m come out 15–18 m too low. One full GeoTIFF scene runs end to end in 42 s on an 8 GB laptop.

# 3 DEFINITION OF THE PROBLEM

## 3.1 Problem statement

> Estimate a reliable height for every pixel of a single optical satellite image, in metres, and show how far those heights can be trusted.

Disaster managers need the height of buildings, trees and ground within hours of an event, but most of India has no recent 3D survey. Optical satellite images exist for almost every place and date, yet one image is flat: it shows what is there, not how tall it is.

## 3.2 Origin and identification of the problem

The problem comes from ISRO, which set it as SIH26175 in Smart India Hackathon 2026. Behind it is a practical gap. India's disaster managers lack up-to-date, high-resolution height data. Airborne LiDAR is costly and rare. Stereo satellite pairs, which give height directly, are not always available for the place and date of an event. Free 30 m terrain models miss individual buildings. The statement therefore asks for:

- a relative surface model (DSM) from a PNG or JPG image, and an absolute DSM in metres above sea level from a GeoTIFF;
- scale calibration from a low-resolution terrain model (DEM) or a few ground control points (GCPs);
- a 3D flythrough in which users check heights against reference data using RMSE, MAE and correlation;
- results that hold across urban, sparse, hilly and forested land;
- a standalone, offline tool with documentation.

## 3.3 Methodology adopted to identify the problem

The problem reached us as a defined statement, so our work went into finding where the real gap lies. We used four methods:

1. **Study of the problem statement.** We split SIH26175 into 13 testable requirements and traced each one to a feature (README requirement table).
2. **Literature review** of monocular height estimation, global terrain models (Copernicus GLO-30, FABDEM, SRTM, CartoDEM) and laser missions (ICESat-2, GEDI).
3. **Discussions** within the team and with mentors, to decide what a disaster manager needs from a height map.
4. **Benchmarking existing DEMs against NASA laser data.** Before building anything, we measured how far the free terrain models sit from ICESat-2 ground points in Indian towns, hills and forests.

We did not run surveys, key-informant interviews or focus-group discussions. [TEAM TO FILL: who we spoke to and what we learned]

## 3.4 Detailed description of the identified problem

Three gaps make single-image height hard to use in practice.

- **Height from one image is ambiguous.** From above, a tall building and a short one can look alike. Models learn cues such as shadows and roof size, and they under-estimate tall objects. On our tests, objects above 20 m come out 15–18 m too low.
- **Height above ground is not height above sea level.** Flood and drainage analysis needs absolute elevation. Free terrain models are 30 m per pixel, and in towns and forests they include roofs and canopy, so their "ground" is too high. In our two urban Indian test scenes, raw GLO-30 sits 5.28 m from laser-measured ground on average.
- **No one can tell how wrong the map is.** Most tools give a height map with no error figure. A relief officer cannot tell whether a 2 m flood depth is good to half a metre or to five.

The tool must also work across India's very different landscapes, from dense cities to hill towns and river banks (Figure 1).



Delhi (urban) | Mumbai (high-rise) | Shimla (hill town)

Guwahati (riverside) | Chandigarh (planned city) | IIT Ropar (campus)

Figure 1. Demo inputs across Indian landscapes. Esri World Imagery, for demonstration only.

## 3.5 Current developments in the domain

- **Monocular depth models.** Vision transformers with dense-prediction heads (DPT) now estimate depth from a single image. Remote-sensing versions predict height above ground instead of camera distance.

- **RS3DAda and SynRS3D.** Song et al. (NeurIPS 2024) trained RS3DAda on SynRS3D, a large synthetic dataset of rendered cities, and adapted it to real images without labels (Figure 2). It is the height model we use.
- **DINOv2.** Meta's self-supervised image encoder gives general features that transfer well to aerial and satellite images. RS3DAda uses its ViT-L/14 version.
- **Global terrain models.** Copernicus GLO-30 is free worldwide at 30 m. FABDEM removes buildings and forest from it, but its licence is non-commercial.
- **Satellite lasers.** NASA's ICESat-2 (ATL03 photons, ATL08 ground and canopy classes) and GEDI give accurate heights along tracks across India.
- **Indian sources.** Cartosat imagery and CartoDEM from NRSC exist, but access is login-gated, so they are not yet tested here.

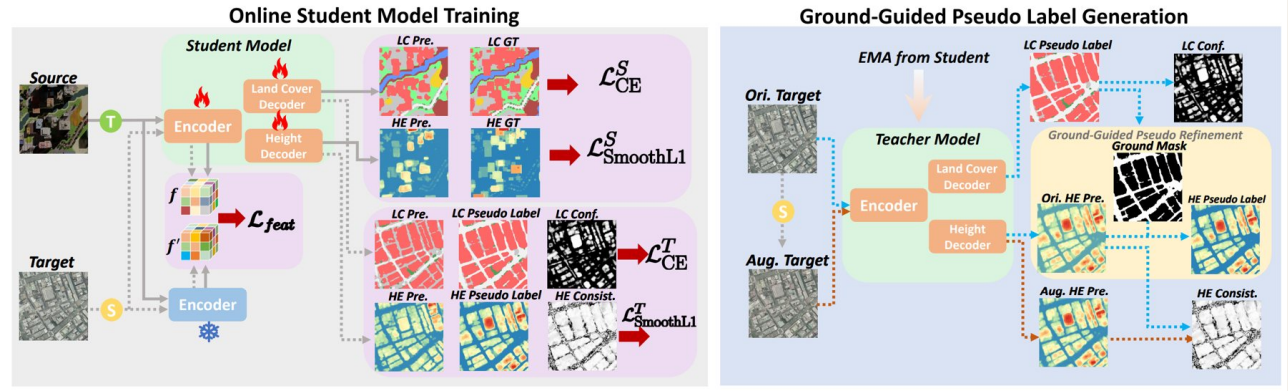



Figure 2. RS3DAda architecture (Song et al., NeurIPS 2024), the height model DepthWizard uses. Figure from the paper, arXiv:2406.18151.

## 3.6 Need and significance of resolving the problem

A trustworthy height map from one image would let disaster teams:

- estimate which streets and homes a given water rise would reach, before the flood;
- route rescue teams away from deep water and steep ground;
- pick safe helipad and relief-camp sites;
- count affected buildings after the event.

Today these tasks rely on coarse 30 m terrain models or are not done at all. Closing that gap, and saying how accurate the result is, makes the output usable for decisions.

# 4 AIMS, OBJECTIVES AND GOALS

> The aim is one offline tool that turns a single satellite image into a height map that disaster teams can use and check.

Each objective closes one of the gaps in section 3.4 and has a test that shows whether it was met.

**Table 1.** Objectives and how each is measured.

| # | Objective | Gap it closes | Measure of success |
|---|---|---|---|
| O1 | Estimate height above ground for every pixel of one RGB image | Height is ambiguous from one image | Lower MAE than a flat-ground baseline on unseen LiDAR and laser data |
| O2 | Convert it to an absolute DSM in metres above sea level | Height above ground is not elevation | Terrain error against held-out ICESat-2 ground; better than raw GLO-30 in towns |
| O3 | Calibrate scale with a coarse DEM or a few ground control points | Model output has no fixed scale | A leave-one-out error reported with every calibration |
| O4 | Validate heights inside the app by landscape | No one can tell how wrong the map is | MAE, RMSE, bias and r per scene and per landscape |
| O5 | Attach an honest uncertainty to building heights | No error figure on outputs | A 90% interval whose coverage is measured on held-out cities |
| O6 | Show results as a 3D flythrough with disaster tools | Results must be usable, not only accurate | Prepare, Respond, Relief and Assess tools working on any processed scene |
| O7 | Run offline on an ordinary laptop | Field offices lack GPUs and internet | One-click install, CPU fallback, end-to-end time measured on a laptop |

# 5 TOOLS AND TECHNIQUES (INTERVENTIONS)

> DepthWizard combines a published deep-learning height model with free global elevation data and satellite lasers, inside one local app.

**Table 2.** Tools and techniques, and the role of each.

| Part | Tool or technique | Role |
|---|---|---|
| Height model | RS3DAda: DINOv2 ViT-L/14 encoder + DPT heads, plus an 8-class land-cover head; released weights, no fine-tuning | Height above ground and land cover for every pixel |
| Inference | Tiles padded to 518 px with overlap, Gaussian blending, 4-flip test-time augmentation | Seamless maps of large scenes; the flip spread becomes a “consistency score” |
| Terrain base | Copernicus GLO-30 DEM (or CartoDEM / SRTM placed locally) | Ground elevation, with roofs and canopy removed using the model’s own heights and land cover |
| Laser levelling | NASA ICESat-2 ATL03 / ATL08 ground photons via SlideRule | Shifts the terrain to laser-measured ground; refused when only one laser pass crosses the scene |
| Roof filter (N5) | Drops “ground” photons that sit on a building and over 3 m above the other ground photons | Stops rooftops being used as ground |
| GCP calibration | Least-squares fit: 1 point sets the level, 2+ the scale, 4+ the tilt | Absolute heights from a few surveyed points, even for PNG/JPG |
| Uncertainty | Split-conformal 90% intervals calibrated on GAMUS LiDAR | An error range on each building block’s height |
| Validation | MAE, RMSE, bias, NMAD, Pearson r, share within ±2 m; bootstrap 95% CIs | The ISRO metrics, in the app and in the benchmarks |
| Viewer | React / TypeScript front end with three.js; Python local server | Orbit, free-fly and street-level cameras, drawn-path flythrough, video, GeoTIFF / GLB export |
| Disaster tools | Connected-flood fill, rain and river timeline (Open-Meteo), A* routing, slope classes (BIS IS 14496), helipad and camp siting (ICAO clearance rule of thumb, Sphere camp standard) | Before, during, shelter and after-event planning |

We chose RS3DAda for three reasons. It was trained for remote sensing, so it predicts height above ground rather than distance from a camera. It predicts land cover in the same pass, which the terrain step needs to find roofs and trees. And its weights are public, so anyone can reproduce our numbers. We did not fine-tune it: every test in this report uses the same released weights.




**Figure 3.** Left: Explore mode, a 3D reconstruction of Bengaluru Electronic City from one image, with fly-through and GeoTIFF / GLB export. Right: Validate & calibrate mode, comparing heights with LiDAR, survey points or ICESat-2, and calibrating with ground control points.

# 6 DETAILED WORK PLAN AND TECHNOLOGICAL INTERVENTIONS

> A scene passes through seven steps, from one uploaded image to a checked surface model in 3D.

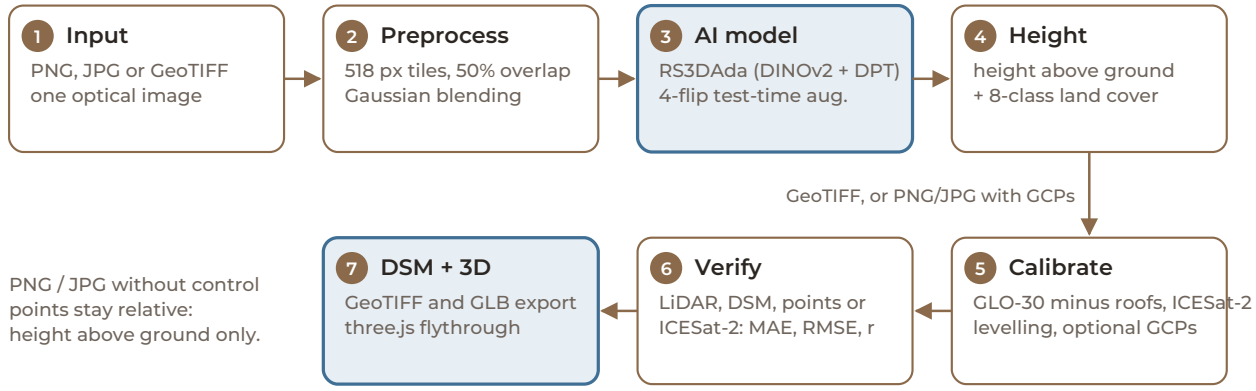



Figure 4. The DepthWizard pipeline. PNG and JPG images stop at relative heights unless ground control points are added; GeoTIFFs go on to metres above sea level.

1. **Input.** The user picks a scene or uploads a PNG, JPG or GeoTIFF (Figure 5).
2. **Preprocess.** The image is cut into overlapping tiles so scenes of any size fit the model.
3. **AI model.** RS3DAda predicts each tile four times (original and flipped) and averages the results.
4. **Height.** The tiles are blended into one height-above-ground map with a land-cover map.
5. **Calibrate.** For a GeoTIFF, the app fetches GLO-30, removes roofs and trees, and levels the terrain to ICESat-2 ground photons when two or more laser passes cross the scene. Ground control points, if given, refine the result.
6. **Verify.** The user compares heights with a reference raster, surveyed points or ICESat-2, and gets MAE, RMSE, bias and r by landscape.
7. **DSM + 3D.** The absolute DSM is shown as terrain you can fly through, and exported as GeoTIFF or GLB.

## How the terrain is cleaned and levelled

Step 5 is where most of the engineering went. GLO-30 is a surface model, so in towns and forests its “ground” includes roofs and canopy. For each 30 m cell we subtract the model’s mean height above ground, and trust the cell in proportion to how much of it the land-cover map calls open ground. Cells covered by roofs or trees are filled from their neighbours.

The cleaned terrain is then compared with ICESat-2 photons that NASA classes as ground, and shifted by the median difference. With only one laser pass the app refuses, because a

single line can carry a local DEM error into the whole scene. The N5 rule also drops “ground” photons that sit on a building and more than 3 m above the other ground photons.

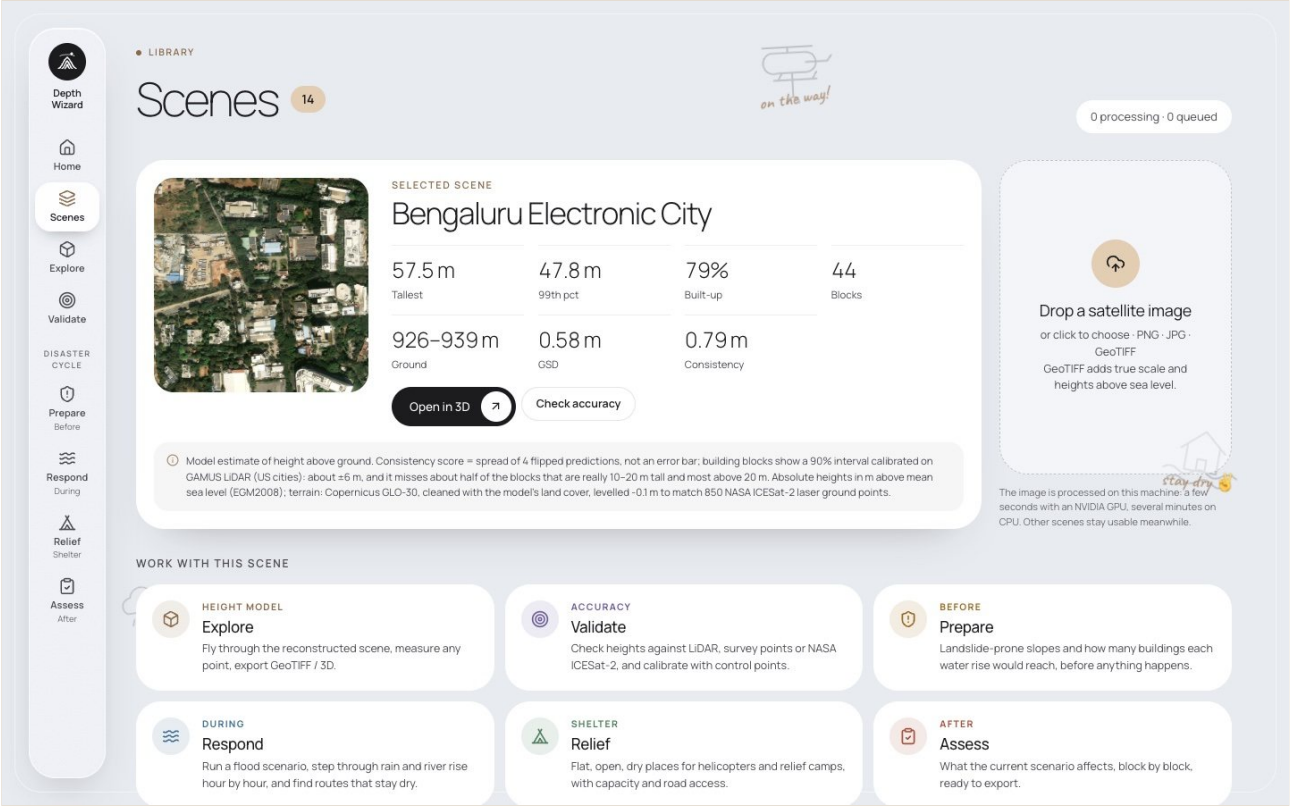



Figure 5. Scene library: choose a demo scene or upload a PNG, JPG or GeoTIFF.

## Timeline

The project moved from a working model to a measured, documented app in October 2026. Earlier work, before the first dated update, covered the problem study, the choice of model and the first versions of the deck.

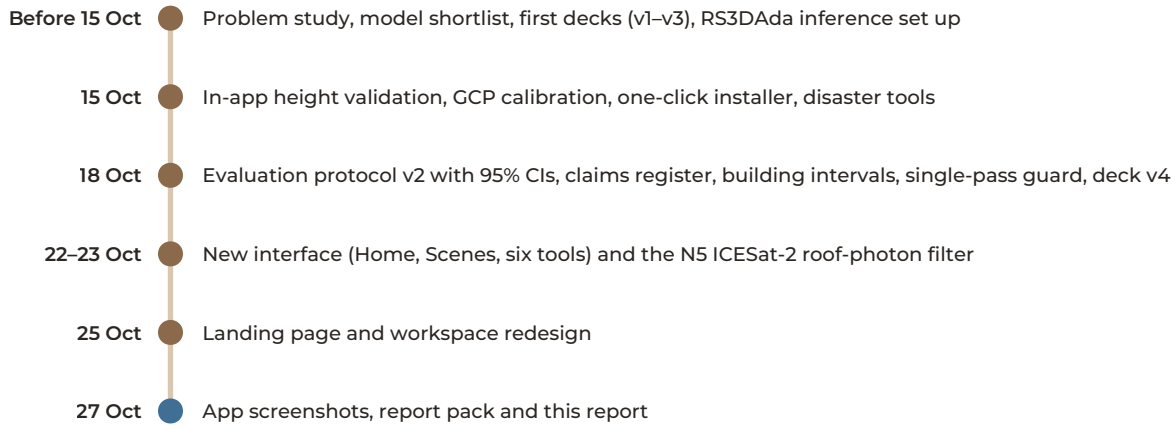



Figure 6. Project timeline, from the team’s dated update notes and repository history.

## Disaster tools across the disaster cycle

The four tools follow the disaster cycle (Figure 7). Each is labelled in the app as a scenario, not a forecast.

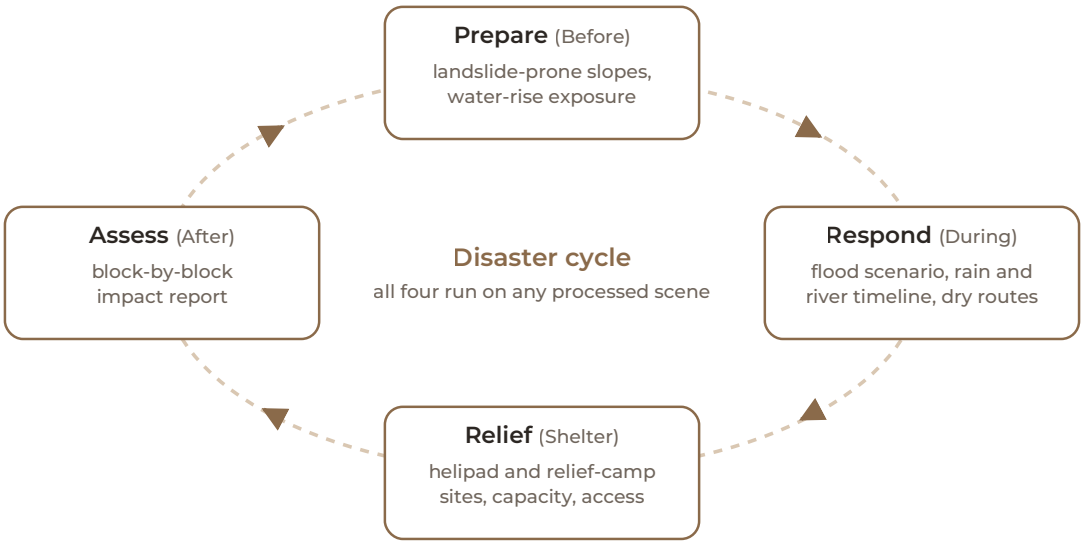
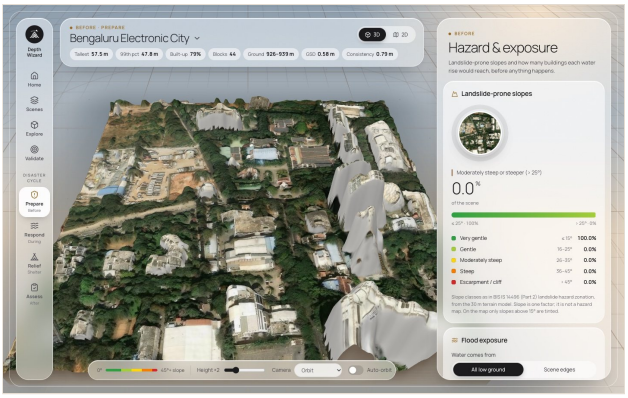
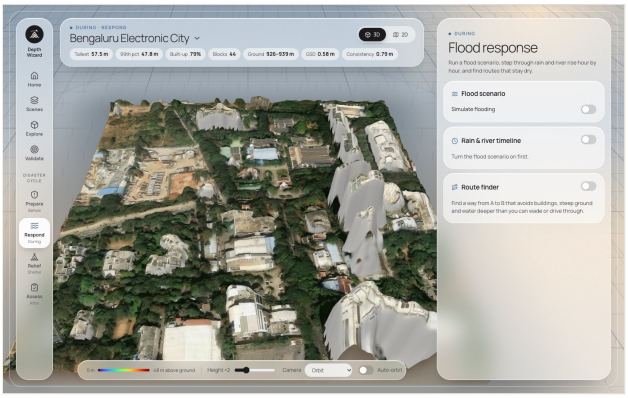


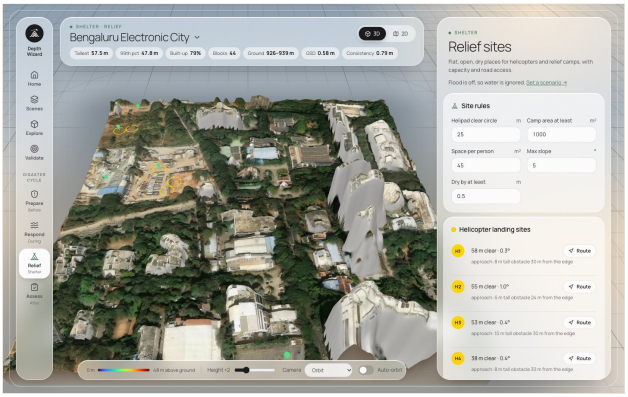

**Figure 7.** The disaster cycle as DepthWizard's four tools: Prepare, Respond, Relief and Assess.



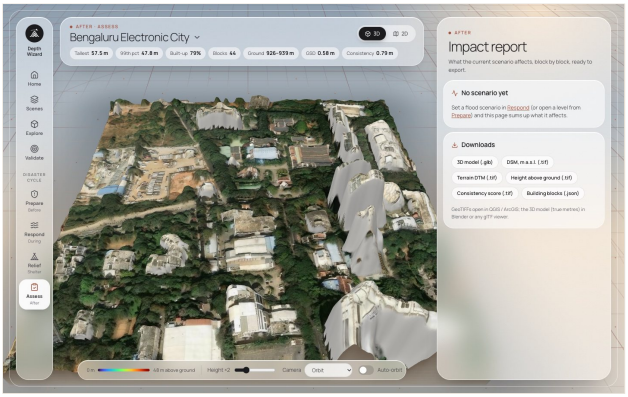

Prepare: hazard and exposure




Respond: flood response




Relief: relief sites




Assess: impact report

**Figure 8.** The four tools. Prepare shows landslide-prone slopes and how many buildings each water rise would reach. Respond runs a flood scenario hour by hour and finds routes that avoid buildings, steep slopes and deep water. Relief finds flat, open, dry sites for helipads and relief camps. Assess reports what the scenario affects, block by block.

# 7 NOVELTY AND INNOVATION OF THE PROPOSED INTERVENTIONS

> The height model is published work; what is new is how DepthWizard turns its output into numbers people can trust and act on.

- **Laser-levelled absolute heights from one image.** Most single-image tools stop at relative height. We strip roofs and canopy from GLO-30 using the model's own output, then level the result to ICESat-2 ground photons. In our two urban Indian scenes this terrain is 2.36 m from laser ground, against 5.28 m for raw GLO-30 and 2.80 m for FABDEM.
- **Safety rules learned from failures.** The app refuses to level from a single laser pass. In Silent Valley one pass along a forested valley floor would have shifted the whole scene by 46.59 m. The N5 roof filter stops ICESat-2 photons labelled "ground" on rooftops from being used as ground.
- **Honest uncertainty.** Building blocks carry a 90% interval with measured coverage. The per-pixel flip spread is labelled a "consistency score", not an error bar, because it ranks errors on flat ground (Spearman 0.44) but not on tall objects (−0.03).
- **Validation inside the tool.** Users upload a reference raster or surveyed points, or fetch ICESat-2, and get the ISRO metrics by landscape. For ICESat-2, the passes used for levelling are never used for checking.
- **A claims register.** Every number in this report, the README and the deck links to the result file it came from, with a 95% confidence interval and a baseline at the same points.

# 8 APPROACHES TO IMPLEMENT THE INTERVENTION PLANS

DepthWizard can reach users in four ways, from a single laptop to national platforms.

- **Standalone offline app for field teams.** A district or state office installs it with one click and runs it offline. Internet is needed only to fetch terrain, laser points and rain forecasts for a new scene. One full GeoTIFF scene took 42 s on an 8 GB Apple M3 laptop.
- **Integration with NDMA / SDMA GIS workflows.** The outputs are standard GeoTIFF and GLB files, so they load into QGIS, ArcGIS, Bhuvan or a state GIS without the app.
- **Cartosat input.** India's own imagery and CartoDEM would replace the commercial demo basemaps. A data request to NRSC is drafted but not yet sent.
- **A distilled small model for laptops (planned).** A smaller student model, trained to copy RS3DAda and exported to ONNX for fast CPU use, is planned. It is not yet part of the app, and its accuracy will be reported only once measured.

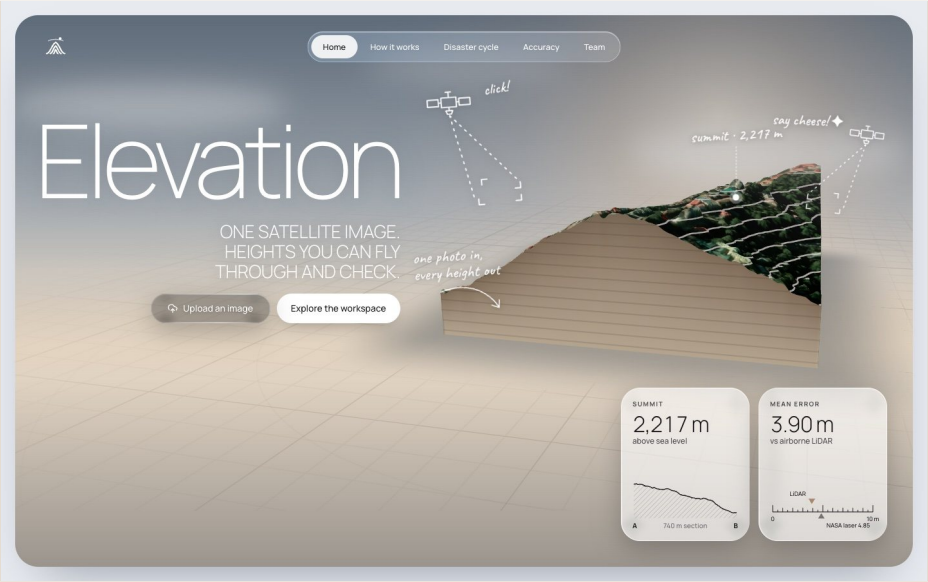



Figure 9. DepthWizard landing page: one satellite image in, heights you can fly through and check.

Each rollout would follow the same steps:

1. Validate on Cartosat imagery and CartoDEM for a few Indian districts.
2. Pilot with one disaster-management office during a monsoon season, comparing flood-reach maps with observed extents.
3. Train staff with the user guide and a short hands-on session.
4. Collect feedback and errors through the in-app validation results.

# 9 POSSIBLE CONSTRAINTS, BARRIERS AND DESIGN ISSUES

> The largest barrier is accuracy on tall objects; the rest are data, hardware and licence limits.

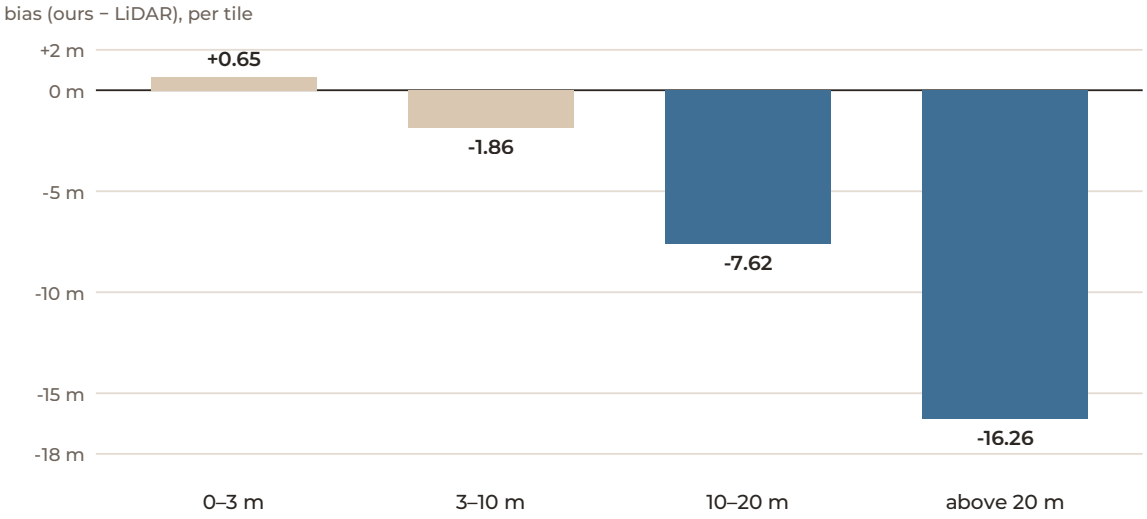



**Figure 10.** Height bias by true (LiDAR) height on GAMUS, 150 tiles. Low objects are about right; tall ones come out far too low. The same pattern appears on satellite tiles and against ICESat-2 in India.

**Table 3.** Constraints, their effect, and how DepthWizard handles them now.

| Constraint | Effect | Handling now |
|---|---|---|
| Tall objects come out too low: −16.26 m above 20 m on GAMUS, −16.80 m (Atlanta), −15.25 m (Buenos Aires), −18.31 m in India | High-rise refuge heights and debris volumes are under-estimated | Stated on every result; fixes listed as future work |
| Tree canopy predicted at 0.18–0.60 of its true height (GAMUS, mostly leaf-off) | Forest heights and flood shelter under canopy are unreliable | Forest results flagged in the app |
| Absolute DSM not better than the raw Copernicus DSM at laser footprints (9.14 vs 8.06 m) | The gain is detail, not average accuracy | Reported openly with the baseline |
| Building intervals miss tall buildings: 49–67% coverage at 10–20 m in DC and NYC, 14–30% above 20 m | The ± on tall buildings is too narrow | The viewer warns that the interval misses most buildings over 10 m |
| Forest terrain is about 20 m off for every DEM (ours 21.99 m, the worst) | Flood and slope results under forest are unreliable | Flagged; better forest terrain is future work |
| Cartosat and CartoDEM are login-gated | India's own imagery is untested | Data request to NRSC drafted |
| Non-commercial licences on the SynRS3D training data, FABDEM and the demo imagery | Commercial use needs new weights or permissions | Research and government pilots only |

| | | |
|---|---|---|
| Development on an 8 GB laptop | One model process at a time; no fine-tuning | Released weights used as they are; the pipeline is tuned to fit in memory |




**Figure 11.** Trees are predicted at a fraction of their true height (leaf-off GAMUS tiles).

The reference data has limits of its own. In dense towns, NASA's ground class can include rooftops: in Chandigarh every one of the 35 held-out "ground" photons was a roof. The N5 filter now removes such photons, so Chandigarh has no laser check until a pass with real ground is available. Most laser passes are also years away from the image date, and India has only two test scenes per landscape.

## Design issues

- **Offline first.** Heavy data (terrain, laser photons) is cached per scene, so a scene opened once works without internet.
- **Honest labels.** Disaster outputs are labelled scenarios, not forecasts, and the flip spread is not presented as an error bar.
- **Speed against accuracy.** The full ViT-L model is the most accurate but slow on CPU. A faster model must be measured on the same tests before it can replace it.

# 10 EXPERTISE AVAILABLE WITH EACH STUDENT

The project needed four kinds of skill: machine learning, geospatial analysis, front-end development and evaluation. The rows below are for the team to complete.

Table 4. Expertise of each student.

| Student | Machine learning | Geospatial | Front end | Evaluation |
|---|---|---|---|---|
| Himanshu | [TEAM TO FILL] | [TEAM TO FILL] | [TEAM TO FILL] | [TEAM TO FILL] |
| Shlok | [TEAM TO FILL] | [TEAM TO FILL] | [TEAM TO FILL] | [TEAM TO FILL] |

# 11 EXPECTED OUTCOMES

> The main outcome is an offline tool that gives a checked height map from one image. It beats a naive baseline on every height test and matches the published model on satellite imagery.

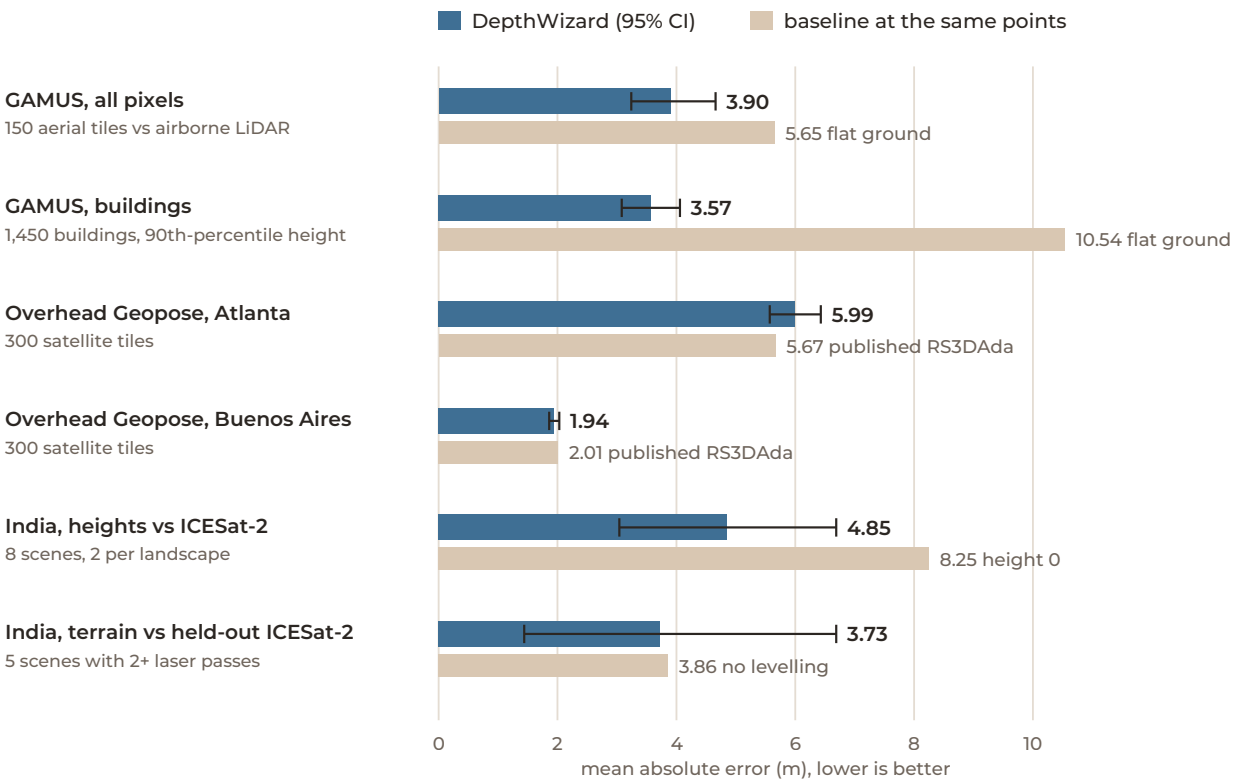



Figure 12. DepthWizard against a baseline measured at the same points. All tests are zero-shot. GAMUS is aerial imagery (0.33 m), not satellite; the satellite claim rests on Overhead Geopose, where the published RS3DAda numbers reproduce within our 95% CIs.

Three results stand out. On 1,450 GAMUS buildings the error is 3.57 m, against 10.54 m for a flat-ground guess. On eight Indian scenes the model alone is 4.85 m from ICESat-2 segment heights, against 8.25 m for "height 0". Laser levelling removes the terrain's average offset (bias +2.58 m to −0.27 m) but not its typical error.

The results also show where the tool can be trusted. On GAMUS it is most accurate in urban areas (2.90 m, against 5.66 m for flat ground) and least accurate under trees (6.67 m). On sparse, open ground a flat-ground guess is better (0.55 m against our 0.96 m), because predicting zero is already right there. The absolute DSM adds detail rather than average accuracy. Against GEDI canopy and roof tops it beats the raw GLO-30 DSM (5.97 vs 8.39 m), but against all ICESat-2 photons it does not (9.14 vs 8.06 m).

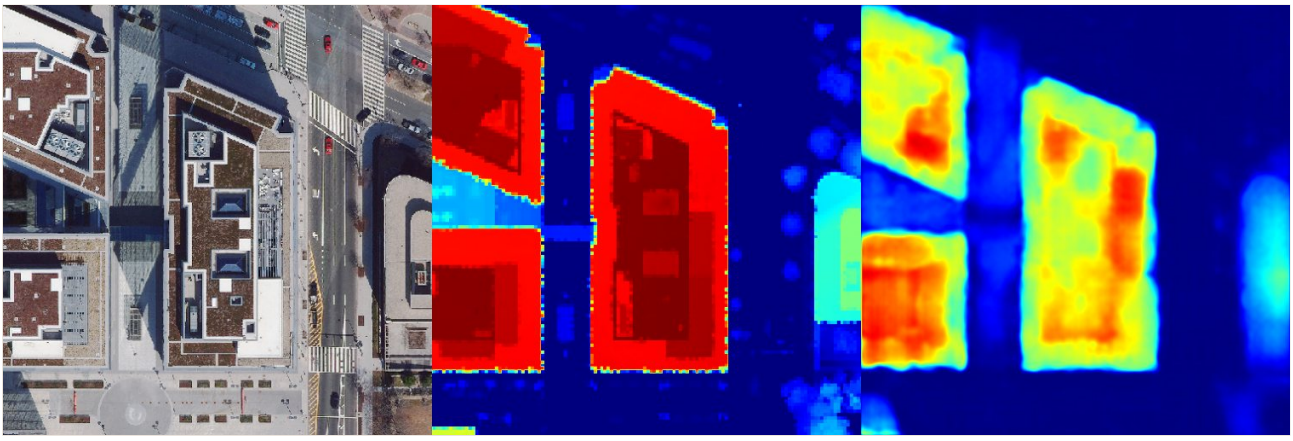


Figure 13. GAMUS tile: aerial image, airborne-LiDAR height, and DepthWizard's predicted height (zero-shot).

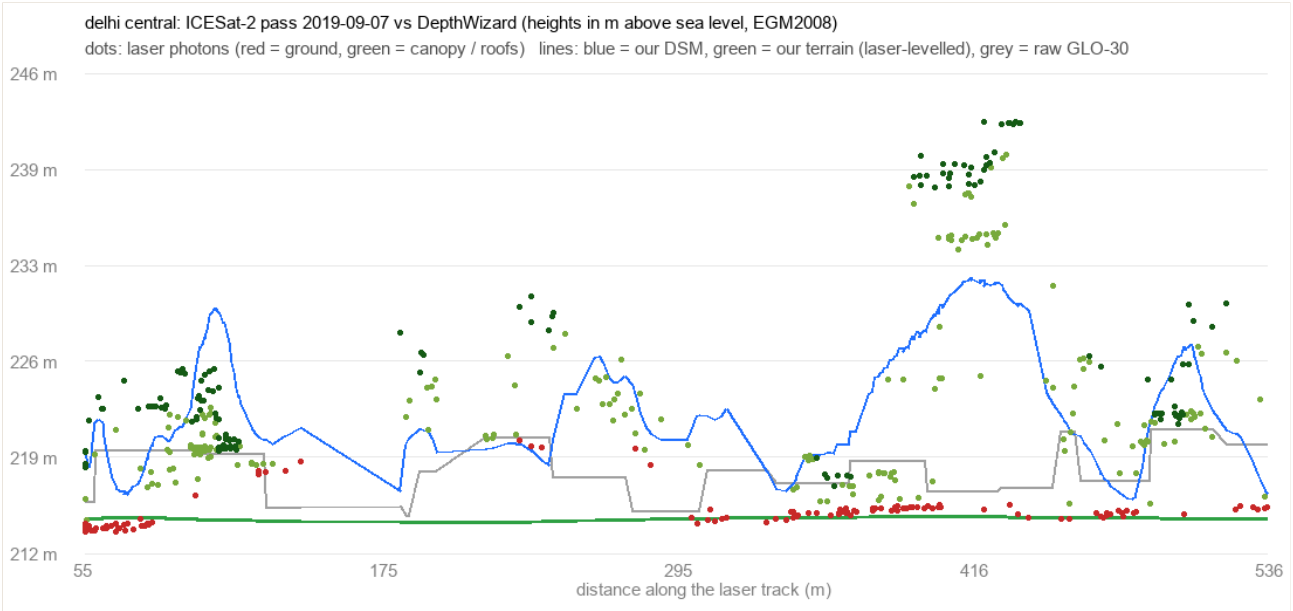



Figure 14. Delhi: heights along an ICESat-2 laser pass. Red = ground photons, green = canopy / roofs; blue = our DSM, green line = our levelled terrain, grey = raw GLO-30.

## Flood scenario test, Porto Alegre, May 2024

To test the flood tool, we flooded the pre-flood image of four small windows and compared the result with the observed extent. With ICESat-2 levelling the agreement (CSI) rises from 0.20 with raw GLO-30 to 0.79. This is a **scenario test, not a forecast**: the windows are small, and one would not be levelled by the current single-pass rule.

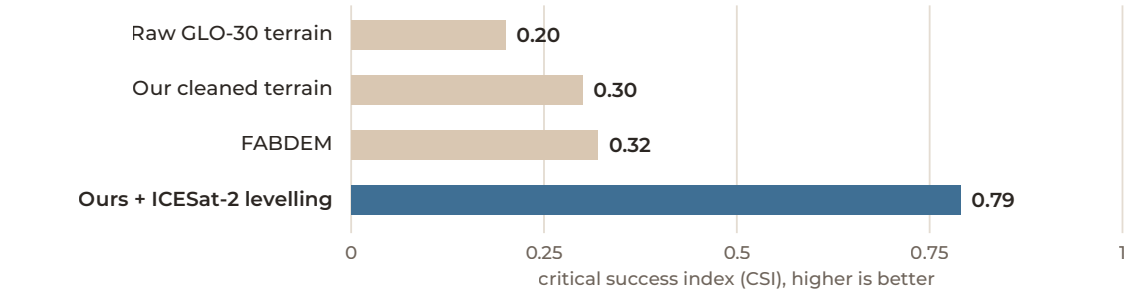



**Figure 15.** Flood agreement with the observed 6 May 2024 extent (IPH/UFRGS), four windows with ICESat-2 photons. Scenario test, not a forecast.




**Figure 16.** Flood scenario test: our flood from the pre-flood image vs the observed extent (blue = both wet, orange = only model wet, red = only observed wet). Imagery: Maxar Open Data (CC BY-NC 4.0). Scenario test, not a forecast.

Beyond these numbers, the expected deliverables are:

- relative and absolute DSM GeoTIFFs for any uploaded scene;
- a 3D flythrough with video export for briefings (cover image);
- per-scene validation reports in the ISRO metrics;
- flood-reach, route and relief-site maps, labelled as scenarios;
- open documentation and a claims register, so others can reproduce each number.

# 12 SUGGESTED PLAN OF ACTION FOR UTILISATION OF THE EXPECTED OUTCOME

> The outcomes should reach users in four stages, each gated on evidence from the one before.

1. **Close the accuracy gaps.**
   - Reduce the tall-object bias, for example with shadow length and sun elevation, a correction per land-cover class, or a canopy-height model for trees, without making low objects worse.
   - Validate on Cartosat imagery and CartoDEM, with more Indian scenes per landscape.
   - Test a drainage-aware flood model (HAND) on one real Indian flood (Assam, Wayanad 2024 or Himachal 2023) against a Sentinel-1 or NRSC flood extent.
   - Measure the planned small student model on the same benchmarks before offering it as a fast mode.
2. **Pilot with one State Disaster Management Authority** for one monsoon season: pre-monsoon flood-reach maps for a few flood-prone districts, compared with observed flood extents.
3. **Hand over to a nodal agency.** Share the code, documentation and claims register with NRSC / ISRO, so the model can run on their GPU servers and feed Bhuvan.
4. **Teach and extend.** Use the tool in geography and planning courses, and invite students to add Indian test scenes.

Beyond disaster management, the same height maps can support urban planning, rooftop-solar studies and checks on unauthorised construction.

# 13 CONCLUSION

> One optical satellite image, combined with free terrain and laser data, can give a usable and checkable height map offline.

DepthWizard beats a flat-ground guess on every height test, reproduces the published model's satellite results, and cuts town terrain error against laser ground from 5.28 m to 2.36 m. It runs a full scene in 42 s on an ordinary laptop.

It is not yet a replacement for a survey. Tall buildings come out 15–18 m too low, forest terrain is about 20 m off, and India's own Cartosat imagery is still untested. Because each result comes with its error and a baseline, users can see where to trust it and where not to. The next steps are fixing the tall-object bias, validating on Indian data, and piloting with one disaster-management office.

# 14 CONTRIBUTION OF EACH STUDENT

**Table 5.** Contribution of each student to the project and this report.

| Student | Contribution to the project | Contribution to this report |
|---|---|---|
| Himanshu | [TEAM TO FILL] | [TEAM TO FILL] |
| Shlok | [TEAM TO FILL] | [TEAM TO FILL] |

Word count of the body text (sections 2–14; tables, figures and captions excluded): 2,826

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# A APPENDIX: FULL ACCURACY TABLES

All values are copied from docs/CLAIMS.md (row IDs in the first column). Mean over tiles or scenes with a 95% bootstrap CI; "pooled" is all pixels or points together. Zero-shot: nothing was trained or tuned on these test sets.

**Table 6.** Height above ground vs airborne LiDAR, GAMUS (aerial 0.33 m imagery; DC and NYC trees leaf-off).

| ID | Measure | n | Value |
|---|---|---|---|
| G1 | Overall MAE, 4-flip TTA | 150 tiles | 3.90 m [3.24, 4.66] (pooled 3.88) |
| G2 | Flat-ground baseline MAE | 150 tiles | 5.65 m [4.88, 6.50] |
| G3 | RMSE / bias / NMAD / r | 150 tiles | RMSE 5.58 [4.82, 6.40]; bias −2.61 [−3.41, −1.87]; NMAD 2.52 [2.09, 3.01]; r 0.61 [0.57, 0.65] |
| G5 | MAE per city | 50 tiles each | DC 5.16 [3.77, 6.75]; NYC 4.69 [3.49, 5.99]; PHL 1.86 [1.44, 2.39] |
| G6 | Pixels with LiDAR height > 3 m | 147 tiles | MAE 6.65 [5.90, 7.44], bias −5.27; baseline 11.10 |
| G7 | Bias by height bin | 150 / 145 / 134 / 76 | 0–3 m +0.65; 3–10 m −1.86; 10–20 m −7.62; > 20 m −16.26 |
| G8 | Buildings, 90th-percentile height | 1,450 buildings | 3.57 m [3.08, 4.06]; bias −1.37; baseline 10.54 |
| G9 | By landscape, ours vs baseline | 137 / 141 / 148 | urban 2.90 vs 5.66; forested 6.67 vs 8.70; sparse / open 0.96 vs 0.55 (baseline better) |
| G11 | Tree canopy, ours / reference height | 121 tree tiles | DC 0.20, NYC 0.18, PHL 0.60 |
| M2 | 4-flip TTA minus single pass, MAE | 150 tiles | −0.064 m [−0.101, −0.035] |

**Table 7.** Satellite imagery, Overhead Geopose (whole-image MAE, authors' evaluation protocol).

| ID | City | n | Ours | Published RS3DAda | Published source-only |
|---|---|---|---|---|---|
| A1 | Atlanta | 300 tiles | 5.99 m [5.57, 6.43] | 5.67 m | 7.26 m |
| A2 | Buenos Aires | 300 tiles | 1.94 m [1.86, 2.03] | 2.01 m | 2.65 m |
| A3 | DepthWizard inference path | 300 each | Atlanta 5.90, Buenos Aires 1.87 | no worse than the authors' script (M3) | |
| A4 | Bias, 10–20 m / > 20 m | 288–300 per bin | Atlanta −8.11 / −16.80; Buenos Aires −5.59 / −15.25 | | |

**Table 8.** Indian scenes vs NASA ICESat-2 and GEDI (8 scenes, 2 per landscape; held-out passes).

| ID | Measure | n | Value |
|---|---|---|---|
| I1 | Height model alone vs ICESat-2 segment heights | 8 scenes | 4.85 m [3.04, 6.69] vs height 0: 8.25; r 0.40 |
| I2 | Height model vs GEDI rh98 | 7 scenes | 9.93 m [4.86, 15.42] vs height 0: 16.89 |
| I3 | Bias, reference > 20 m | 40 segments | −18.31 m |

| | | | |
|---|---|---|---|
| I4 | Terrain vs held-out ground: no levelling / levelled | 5 scenes | 3.86 / 4.02 m; bias +2.58 → −0.27 |
| I5 | Terrain, current app (single-pass guard) | 5 scenes | 3.73 m [1.44, 6.69] |
| I7 | Terrain by landscape: ours / GLO-30 / FABDEM | 2 scenes each | urban 2.36 / 5.28 / 2.80; sparse 1.76 / 1.80 / 1.82; hilly 8.42 / 9.20 / 5.74; forested 21.99 / 20.97 / 19.87 |
| I8 | Absolute DSM vs all ICESat-2 photons: ours vs GLO-30 DSM | 8 scenes | 9.14 vs 8.06 m (ours loses) |
| I9 | Absolute DSM vs GEDI canopy / roof tops | 7 scenes | 5.97 vs 8.39 m |
| I11 | r after removing the terrain plane | 8 scenes | terrain 0.26; surface 0.43 |

**Table 9.** Uncertainty, flood test and what was tried but not shipped.

| ID | Measure | n | Value |
|---|---|---|---|
| U1 | Consistency score vs error (Spearman) | 150 tiles | all pixels 0.44; > 3 m −0.03; trees 0.03 |
| U4 | Building-block 90% interval, held-out city | 959 blocks | DC 86.6%, NYC 83.3%, PHL 97.0%; width about ±6 m |
| U5 | Same, by height | 959 blocks | 10–20 m: DC 67%, NYC 49%, PHL 96%; > 20 m: 14–30% |
| U6 | Interval on Indian roofs | 96 buildings | 71% |
| F1 | Flood CSI: GLO-30 / ours / FABDEM / ours + ICESat-2 | 4 windows | 0.20 / 0.30 / 0.32 / 0.79 (scenario test) |
| F2 | Flood CSI, 7 windows, no levelling: GLO-30 / ours / FABDEM | 7 windows | 0.18 / 0.47 / 0.48 |
| R1 | Isotonic recalibration, held-out city (not shipped) | 3 folds | PHL 1.86 → 2.87 m (worse) |
| T2 | GEDI-densified levelling (not shipped) | 5 scenes | 4.18–4.24 vs 4.02 m |

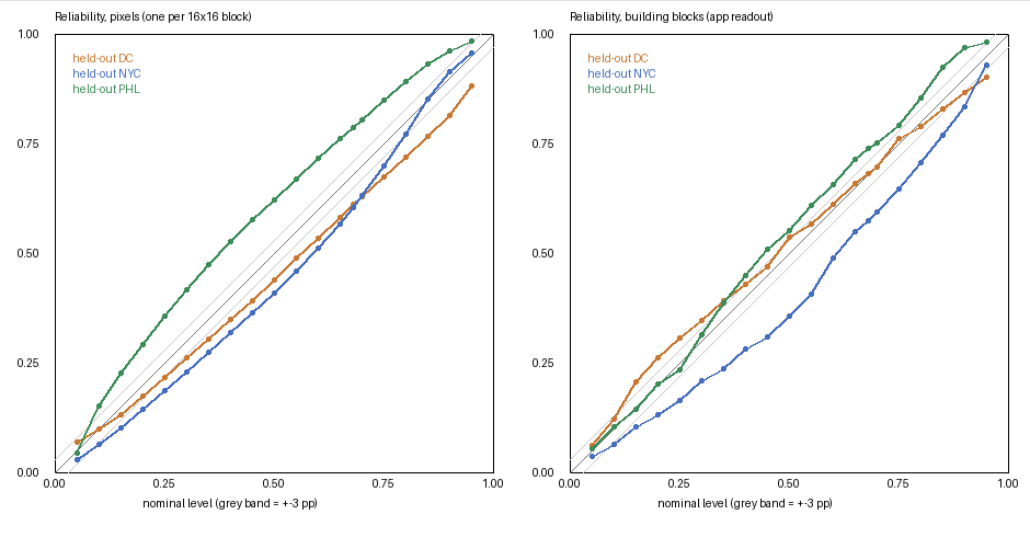



**Figure 17.** Coverage of the 90% building-height interval on held-out cities.

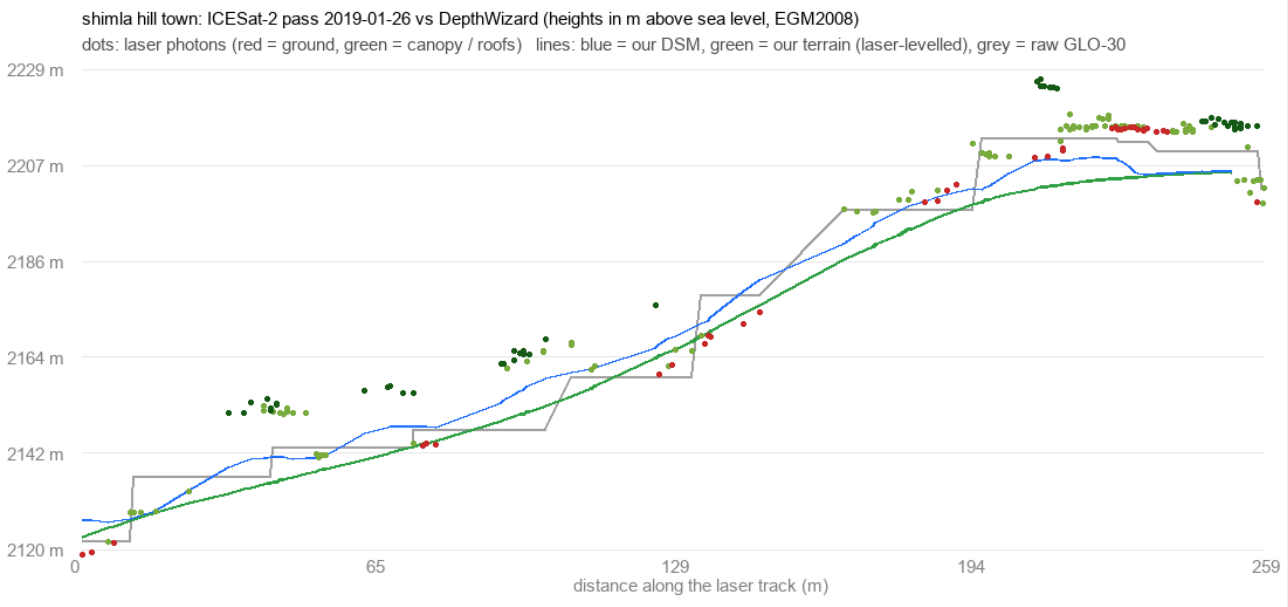



**Figure 18.** Shimla: heights along an ICESat-2 laser pass. Red = ground photons, green = canopy / roofs; blue = our DSM, green line = our levelled terrain, grey = raw GLO-30.

# B APPENDIX: THE DEPTHWIZARD LANDING PAGE

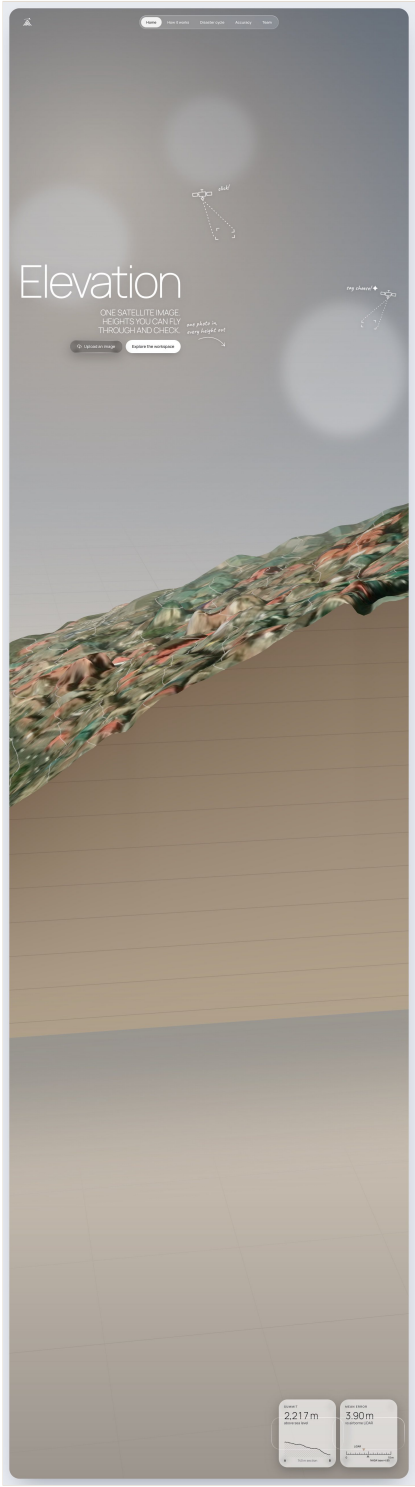



**Figure 19.** The complete DepthWizard landing page.